Module (4)-Assignment

Submitted by: Khansa Rubab
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Materials and Method

Functional Annotation

To investigate the genomic alterations driving Breast cancer, comprehensive annotation was performed on the SRA Run accession SRR33505761whole-genome sequence aligned to human **Chromosome 19.** Reference-guided annotation facilitated the identification and characterization of genomic features, including coding sequences (CDS), genes, and non-coding RNA elements. This detailed mapping was critical for pinpointing Copy Number Variations (CNVs) and Structural Variants (SVs) and for providing functional context to these alterations. The analysis also extended to the prediction of key regulatory regions, such as promoters, to elucidate how these large-scale variants may dysregulate gene expression and contribute to an aggressive disease phenotype.

Results

Genome annotation was performed on the SRA Run accession SRR33505761sample dataset previously aligned to human Chromosome 19. This analysis provided a comprehensive map of the genomic landscape, identifying key functional elements including genes, coding sequences (CDS), and RNA. Additionally, the annotation process included the prediction of gene features such as promoter regions and other regulatory sequences. This foundational data set provides the necessary context for the subsequent identification of CNVs and SVs within these functional elements, directly supporting our investigation into the molecular drivers of this disease. The resulting annotated features, summarized in **Table 1**, serve as a critical foundation for all subsequent analyses using *Geneious version 9.1.*

Table 1: Summary of Annotated Genomic Features from SRA Run Accession SRR33505761 Aligned to Human Chromosome 19

Chromosomal Features	SRR33505761
CDS	7,522
Centromere	1
Exon	221
gene	2,641
Mature Peptide	477
Misc Feature	22,871
Misc_Recomb	2
Misc-RNA	761
Mobile Element	7
mRNA	7,522
ncRNA	1,098
Precursor _RNA	142
Protein _Bind	92
Regulatory	7,071
Repeat Region	6

Signal peptide	621
source	1
Transit _peptide	42
tRNA	(7)

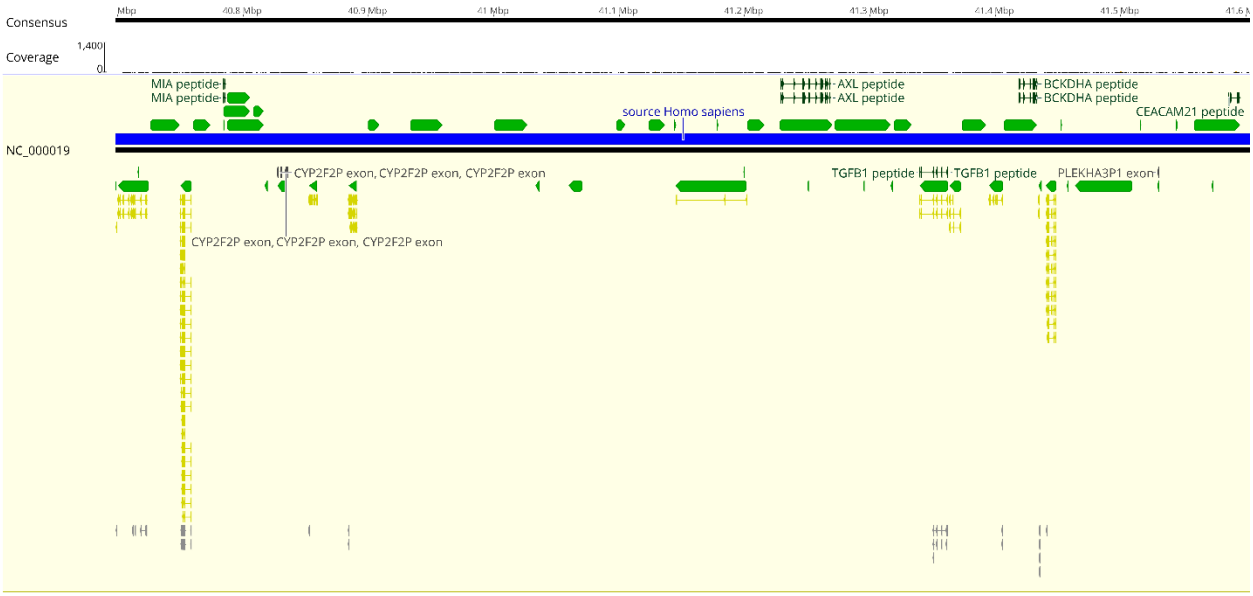


Figure 2: Representative Genomic View of an Annotated Breast Cancer Sample SRR33505761Aligned **to Human Chromosome 19.**